Rev–02 final board. Sensor sheets are carried over unchanged from Rev–01 (origin/adrian, 44ecbd0).
The MCU and Power/M.2 sheets are new for Rev–02. UART and trigger pins on M.2 still need Psionic's agreement.

Accel & Gyro

CS_ADXL
CS_ICM
ADXL_INT0
ADXL_INT1
ICM_INT1
SCLK
MOSI
MISO
FSYNC_ADXL
FSYNC_ICM

CS_ADXL
CS_ICM
INT0_ADXL
INT1_ADXL
INT1_ICM
SCL_Gyro_Accel
SDI_Gyro_Accel
SDO_Gyro_Accel
FSYNC_ADXL
FSYNC_ICM

File: Accel_Gyro.kicad_sch

MCU

File: MCU.kicad_sch

Power_M2_IO

File: Power_M2_IO.kicad_sch

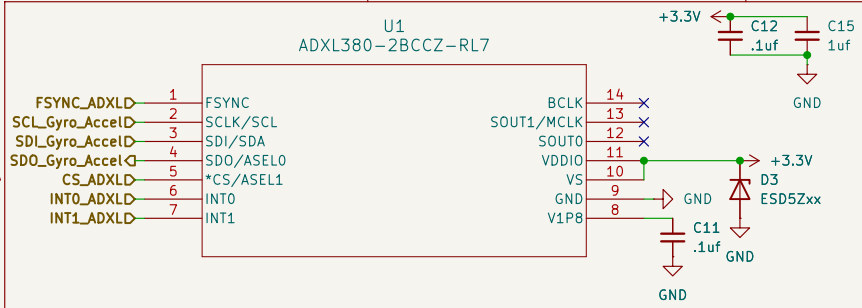
Magnetmeter and Barometer

CS_BARO
Baro_INT
SCLK
MOSI
MISO
CS_MAG
Mag_INT

Baro_CS
Baro_INT
Mag_Baro_SCK
Mag_Baro_SDI
Mag_Baro_SDO
Mag_CS
Mag_INT

File: Mag_Baro.kicad_sch

VT ECE 4805 / Psionic		
Sheet: / File: Final_Board.kicad_sch		
Title: Psionic IMU – Final Board		
Size: A4	Date:	**Rev: 02**
KiCad E.D.A. 10.0.6		Id: 1/5



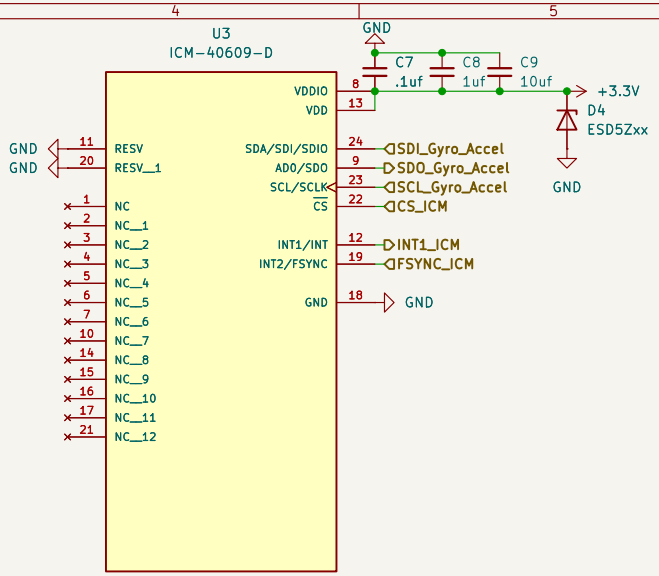


Table 9. Pin Function Descriptions

		Description			
Pin No.	**Mnemonic**	**SPI**	**I²S/TDM**	**I²C**	**PDM**
1	FSYNC	Interrupt Input in FIFO Trigger Mode (Optional)[1]	Frame Sync (FSYNC)	Interrupt Input in FIFO Trigger Mode (Optional)[1]	SOUT1
2	SCLK/SCL	SPI, Clock (SCLK)	Not Applicable	I²C, Serial Clock (SCL)	Not Applicable
3	SDI/SDA	SPI, Serial Data Input (SDI)	Not Applicable	I²C, Serial Data (SDA)	Not Applicable
4	SDO/ASEL0	SPI, Serial Data Output (SDO)	Not Applicable	I²C, Address Select 0 (ASEL0)	Not Applicable
5	$\overline{CS}$/ASEL1	SPI, Chip Select[2]	Not Applicable	I²C, Address Select 1 (ASEL1)	Not Applicable
6	INT0	Interrupt 0	Interrupt 0	Interrupt 0	Interrupt 0
7	INT1	Interrupt 1[1]	Interrupt 1	Interrupt 1[1]	Interrupt 1
8	V_{1P8}	Internally Regulated Voltage (External 1.8 V When V_S Grounded)	Internally Regulated Voltage (External 1.8 V When V_S Grounded)	Internally Regulated Voltage (External 1.8 V When V_S Grounded)	Internally Regulated Voltage (External 1.8 V When V_S Grounded)
9	GND	Ground	Ground	Ground	Ground
10	V_S	Supply Voltage	Supply Voltage	Supply Voltage	Supply Voltage
11	V_{DDIO}	Input and Output Supply	Input and Output Supply	Input and Output Supply	Input and Output Supply
12	S_{OUT0}	Not Applicable[2, 3]	Data Channel 0 (S_{OUT0})	Not Applicable[3]	Data Channel 0 (S_{OUT0})
13	S_{OUT1}/ MCLK	Not Applicable[4]	MCLK or Data Channel 1 (S_{OUT1})	Not Applicable[4]	Not Applicable
14	BCLK	Not Applicable[4]	Bit Clock (BCLK)	Not Applicable[4]	Bit Clock (BCLK)

4 APPLICATIONS INFORMATION

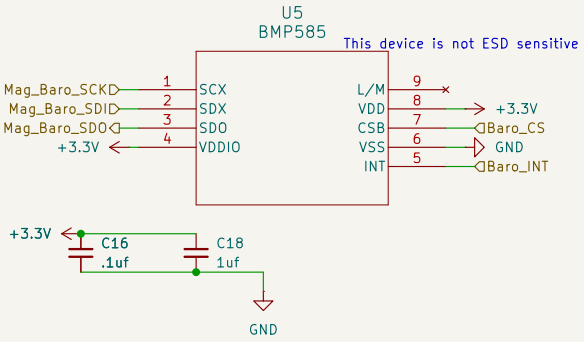
4.1 PIN OUT DIAGRAM AND SIGNAL DESCRIPTION

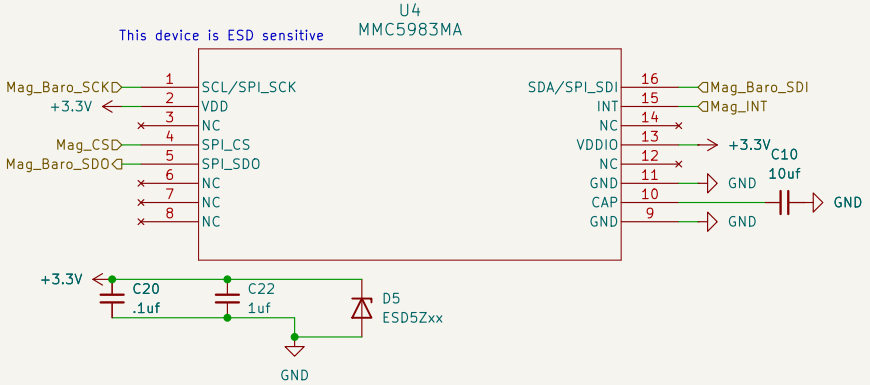
PIN NUMBER	PIN NAME	PIN DESCRIPTION
8	VDDIO	Digital I/O supply voltage
9	AD0 / SDO	AD0: I²C Slave Address LSB SDO: SPI serial data output (4-wire mode)
11	RESV	RESV: Connect to GND
12	INT1 / INT	INT1: Interrupt 1 INT: Interrupt if all interrupts mapped to pin 12
13	VDD	Power supply voltage
18	GND	Power supply ground
19	INT2 / FSYNC	INT2: Interrupt 2 FSYNC: Frame sync input; Connect to GND if FSYNC selected but FSYNC signal not input
20	RESV	Reserved, connect to GND
22	nCS	Chip select (SPI mode only)
23	SCL / SCLK	SCL: I²C serial clock SCLK: SPI serial clock
24	SDA / SDI / SDIO	SDA: I²C serial data SDI: SPI serial data input (4-wire mode) SDIO: SPI serial data I/O (3-wire mode)
1 – 7, 10, 14 – 17, 21	NC	No Connect pins. Do not connect.

Sheet: /Accel & Gyro/
File: Accel_Gyro.kicad_sch

Title:

Size: A4 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 2/5





6.2.2 SPI 4-wire

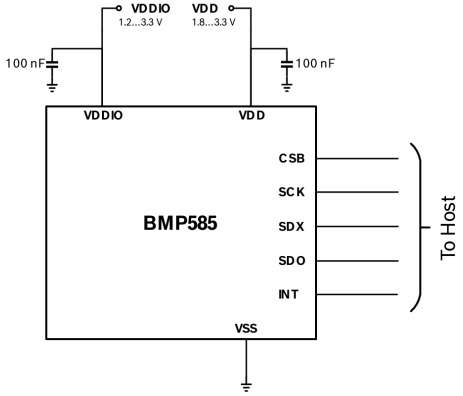
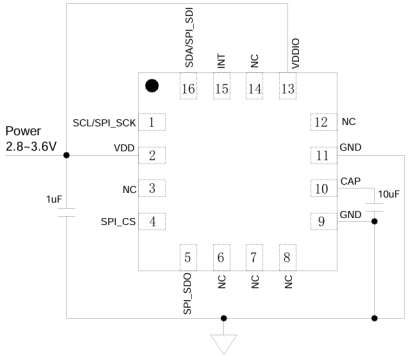


Figure 24: SPI 4-wire connection diagram

Table 26: Pin description

Pin	Name	I/O Type	Description	Connect to	
				SPI 4W	SPI 3W
1	SCX	In	I2C Serial clock SPI Serial clock	SCK	SCK
2	SDX	In/Out	I2C Serial data SPI Serial data input (4-wire) SPI Serial data IO (3-wire)	SDI	SDI/SDO
3	SDO	In/Out	I2C Slave address LSB SPI Serial data output (4-wire)	SDO	DNC
4	VDDIO	Supply	Digital interface power supply	V_{DDIO}	
5	INT	Out	Interrupt or data ready	host INT input, GND[a] or DNC	
6	VSSIO/VSS	Supply	Ground	GND	
7	CSB	In	Chip select for SPI (Low Active)	CSB	CSB
8	VDD	Supply	Analog power supply	V_{DD}	
9	L/M Pad	---	Lasermarking	No external connection possible due to S/R coverage	

a. GND connection is allowed, as long as the IRQ pin is not activated



<Top View>
SPI Single Supply Connection

PIN DESCRIPTIONS

CAP– This is the supply input for SET/RESET coil. MEMSIC recommends a minimum 10 uF low ESR capacitor placed in close proximity to the CAP pin.

GND– This is the ground pin for the magnetic sensor.

SDA/SPI_SDI – This pin is the I²C/SPI serial data line, and operates in I²C FAST (400 KHz)/SPI mode.

SCL/SPI_SCK– This pin is the I²C/SPI serial clock line, and operates in FAST (400 KHz)/SPI mode.

SPI_CS – This pin is chip select line for SPI (active low). Tie to VDDIO for I²C Interface.

SPI_SDO – This pin the SPI serial data out line.

VDD–This pin is the power supply of the ASIC.

VDDIO–This pin is the power supply of the digital interface of the ASIC.

INT– This pin is Interrupt output. Active High. It is held as hi-Z until an interrupt function is enabled.

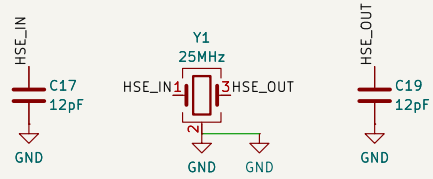
Note: VDD and CAP pins are shorted together inside the device.

Sheet: /Magnetmeter and Barometer/
File: Mag_Baro.kicad_sch

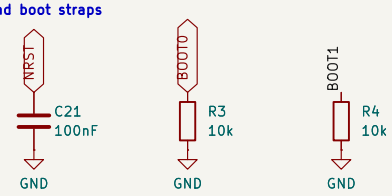
Title:

Size: A4 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 3/5

25 MHz HSE crystal. Set C17 and C19 from the chosen crystal's load capacitance.

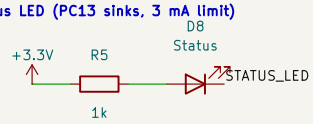


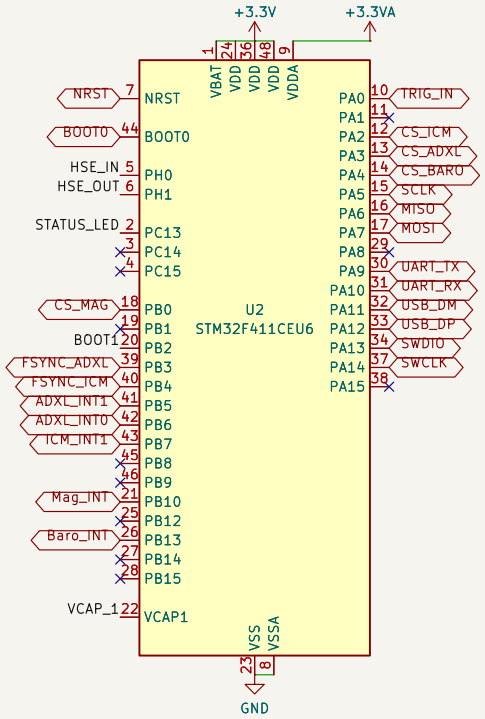
Reset and boot straps



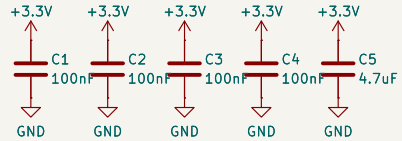
BOOT0 high plus BOOT1 (PB2) low selects the ROM bootloader.

Status LED (PC13 sinks, 3 mA limit)

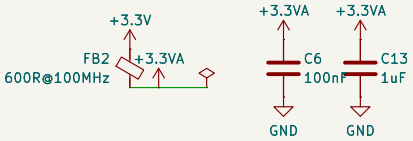




VDD and VBAT decoupling (place at each pin)



VDDA filter



VCAP_1 regulator capacitor (ESR below 1 ohm, per datasheet Table 16)



USB device mode must disable VBUS sensing, because PA9 is used for UART_TX.
Unused pins are marked no-connect; set them to analog in firmware.

VT ECE 4805 / Psionic		
Sheet: /MCU/		
File: MCU.kicad_sch		
Title: MCU: STM32F411CEU6		
Size: A4	Date:	**Rev: 02**
KiCad E.D.A. 10.0.6		Id: 4/5

3.3 V input from the M.2 host

USB protection (route through U6)

UART protection (route through U7)

External trigger input: ESD diode, series resistor, clamps to the rails, pull-down

Test pads (these replace headers J1 and J2 from Rev-01)

M.2 notes: USB uses the standard Key B pins. UART and TRIG_EXT on GPIO_5, GPIO_6 and GPIO_7 are our own choice, to be agreed with Psionic.
CONFIG pins are left open until the host's card-detect needs are known.

VT ECE 4805 / Psionic	
Sheet: /Power_M2_IO/	
File: Power_M2_IO.kicad_sch	
Title: Power, M.2 and I/O	
Size: A4	Date:
KiCad E.D.A. 10.0.6	Rev: 02
	Id: 5/5